College of Engineering and Technology

Project Work/Internship – Student Hand Book

Project Batch ID

368

Degree/ program	B.Tech	**Specialisation**	Computer Science and Engineering
Academic Year	2025-2026 (Even)	**Semester**	8
Course Code	**21CSP401L/ 21CSP402L**	**Course Title**	Project

Name of student	**Register Number**	**Department**	**Mobile Number**	**Email ID**
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Working Title of the Project:	Self-Supervised Contrastive Learning for Universal ECG Representation with Explainable Arrhythmia Detection
Project Site / Location	SRM
Name and address of the company/organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
Name	Dr. P. Akilandeswari	Dr. Ashwini S	
Designation	Associate professor	Assistant professor	
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Problem (or) Product Description:

1. FAILURE OF 1D SIGNAL EXTRACTION IN RURAL CLINICS

Conventional ECG digitization pipelines rely on trace-following algorithms that first segment individual leads from scanned printouts, then reconstruct 1D time-series waveforms. In rural and low-resource clinic environments, this approach fails systematically due to three compounding failure modes:

(a) Optical Degradation: Physical ECG printouts are routinely subject to grid-line interference, thermal-paper fading, scanning artifacts, shadows, and lens glare from smartphone photography. These degrade the signal-to-noise ratio of the extracted 1D trace to clinically unreliable levels.

(b) Grid-Removal Brittleness: Rule-based grid extraction using fixed-frequency band-pass filters assumes uniform print quality. In field conditions, variability in paper stock, ink density, and illumination causes systematic failure of these filters, producing corrupted baseline waveforms.

(c) Infrastructure Dependency: Deploying 1D extraction pipelines requires calibrated flatbed scanners, controlled lighting, and consistent paper formats none of which can be guaranteed in primary health centres or community clinics in Tier-3 regions.

2. THE 2D OPTICAL COMPUTER VISION APPROACH

This project bypasses 1D extraction entirely by treating the raw smartphone photograph of the ECG printout as a native 2D optical image classification task. The system operates on the full 2D image tensor, enabling the encoder to learn clinically discriminative spatial features (morphology of QRS complexes, ST-segment deviation, P-wave abnormalities) directly from pixel intensity patterns.

Preprocessing employs Contrast-Limited Adaptive Histogram Equalization (CLAHE) to normalize local contrast across non-uniform illumination regions, followed by adaptive thresholding to suppress background grid structure and ink bleeding. The cleaned image patches are then fed to a ResNet-18 encoder backbone.

The core training methodology is Self-Supervised Contrastive Learning using the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss. Given a mini-batch of N image samples, augmented view pairs (x_i, x_j) are generated. The NT-Xent loss for a positive pair is: L = -log[exp(sim(z_i,z_j)/tau) / sum_{k!=i} exp(sim(z_i,z_k)/tau)], where sim() denotes cosine similarity and tau is the temperature parameter. This formulation allows the ResNet-18 encoder to learn a universal ECG representation without requiring annotated labels during pretraining.

Post-pretraining, the encoder is fine-tuned on a small labeled subset. Explainability is provided via gradient-based saliency heatmaps (Grad-CAM), which overlay activation regions on P-QRS-T waveform segments, offering physicians visual attribution for each diagnostic decision.

Assumptions and Constraints

TECHNICAL ASSUMPTIONS (Edge Hardware):

A1. Smartphone Camera Resolution: It is assumed that the target deployment device possesses a rear-facing camera with a minimum resolution of 8 megapixels (3264x2448 pixels), sufficient to resolve standard ECG grid spacing (1mm = 0.04 seconds at standard paper speed of 25mm/s) without critical aliasing.

A2. CPU-Only Inference Capability: The inference pipeline is assumed to operate on ARM Cortex-A class processors (e.g., Qualcomm Snapdragon 6xx series or equivalent) without requiring a dedicated GPU or NPU, enabling deployment on commodity Android devices priced under INR 8,000 in the target demographic.

A3. Minimum RAM and Storage: A minimum of 3 GB RAM and 500 MB of available storage is assumed on the deployment device to accommodate the quantized ResNet-18 model weights (INT8 quantized to approximately 12 MB) and the runtime inference framework (TensorFlow Lite or ONNX Runtime).

A4. Ambient Lighting Conditions: The CLAHE preprocessing stage assumes ambient illumination of at least 300 lux (equivalent to standard indoor office lighting), sufficient for the adaptive thresholding algorithm to reliably discriminate ECG trace ink from the background paper grid. Sub-100 lux environments may degrade preprocessing efficacy.

CONSTRAINTS (Training Budget and GPU Requirements):

C1. GPU Compute Budget: The self-supervised pretraining phase (NT-Xent loss, ResNet-18 backbone, batch size 256, 200 epochs) was constrained to a single NVIDIA Tesla T4 GPU (16 GB VRAM) as available on Google Colab Pro+. Scaling to larger architectures (e.g., ResNet-50 or ViT) was beyond the available compute allocation.

C2. Labeled Data Budget: The fine-tuning phase was constrained to operate within a maximum 40% labeled data budget of the available curated ECG image dataset, demonstrating label efficiency as the core research contribution of this work. The 100% labeled fine-tuning scenario serves only as a controlled baseline for overfitting analysis.

C3. Dataset Scale and Scope: Training data is limited to publicly available 12-lead ECG printout imagery and the project-curated smartphone photograph dataset. The model has not been validated on pediatric ECG morphology or ECG printouts from non-standard paper formats (e.g., 50mm/s paper speed), representing a scope constraint on generalization claims.

Stakeholders

Primary Stakeholders: Rural and semi-urban General Practitioners (GPs) and MBBS-level clinicians operating in Primary Health Centres (PHCs) and Community Health Centres (CHCs) with limited specialist cardiology access.

Secondary Stakeholders: ASHA (Accredited Social Health Activist) workers and paramedical staff conducting preventive cardiac screening camps in Tier-2 and Tier-3 districts.

Technical Stakeholders: SRM Institute of Science and Technology Department of Computer Science and Engineering; supervising faculty Dr. P. Akilandeswari; the project research and development team comprising A Lalith Rahul (RA2211003011009) and M V Prahlad Karthik (RA2211003011022).

Institutional / Policy Stakeholders: National Health Mission (NHM) India for potential deployment alignment; digital health initiatives under Ayushman Bharat Digital Mission (ABDM); and international frameworks aligned with UN SDG-3 (Good Health and Well-Being).

Project Outcomes Beneficiaries: Patients in low-resource settings lacking access to in-person cardiologist review, for whom the system provides a first-line AI-assisted triage and diagnostic support mechanism.

Division of work and contributors

Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
15 Jan 2026	19 Jan 2026	Sprint 1 Kickoff: Project requirements review, literature survey on self-supervised contrastive learning and ECG digitization approaches. Repository	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022

		setup (GitHub), environment configuration (Python 3.10, PyTorch 2.1, CUDA 12.1).		
20 Jan 2026	22 Jan 2026	Dataset acquisition and curation: Collection of smartphone-photographed ECG printout images from publicly available repositories (PhysioNet, PTB-XL scans). Organization into arrhythmia classes: Normal Sinus Rhythm, Atrial Fibrillation, Left Bundle Branch Block, Right Bundle Branch Block, ST-Elevation.	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009
23 Jan 2026	25 Jan 2026	Image preprocessing pipeline design: Implementation of CLAHE (Contrast-Limited Adaptive Histogram Equalization) with clip limit 2.0 and tile grid size 8x8 for local contrast normalization across non-uniform illumination regions.	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022
26 Jan 2026	28 Jan 2026	Adaptive thresholding module: Development and validation of adaptive binary thresholding (Gaussian kernel, block size 31) for ECG grid-line suppression. Quantitative evaluation of grid removal using SSIM metric against ground-truth clean traces.	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009
29 Jan 2026	31 Jan 2026	Data augmentation framework: Implementation of contrastive augmentation policy for NT-Xent training pairs - random cropping (224x224), Gaussian blur (sigma 0.1-2.0), random grayscale (p=0.2), color jitter (brightness 0.4, contrast 0.4, saturation 0.4), random horizontal flip (p=0.5).	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022
01 Feb 2026	03 Feb 2026	ResNet-18 backbone integration: Loading pretrained ImageNet weights, modifying final FC layer to output 128-dimensional projection head (MLP: 512->256->128 with ReLU). Verification of forward pass	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009

		with batch size 256, input shape [256,3,224,224].		
04 Feb 2026	06 Feb 2026	NT-Xent Loss implementation: Coding the Normalized Temperature-scaled Cross Entropy loss function with temperature tau=0.07. Validation against published SimCLR benchmark on CIFAR-10 as sanity check. Loss convergence monitoring via TensorBoard.	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022
07 Feb 2026	09 Feb 2026	Pretraining execution (Phase 1): Self-supervised pretraining on full unlabeled ECG image corpus. 200 epochs, Adam optimizer (lr=3e-4, weight decay=1e-4), cosine annealing LR scheduler. Hardware: Google Colab Pro+ single NVIDIA Tesla T4 GPU.	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009
10 Feb 2026	11 Feb 2026	Pretraining evaluation and checkpoint selection: Analysis of NT-Xent loss curves, t-SNE visualization of learned 128-dim embedding space to verify inter-class separability. Selection of best checkpoint by minimum validation loss. Sprint 1 internal review and documentation.	A Lalith Rahul RA2211003011009 M V Prahlad Karthik RA2211003011022	Both

Division of work and contributors

Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
12 Feb 2026	14 Feb 2026	Label budget fine-tuning experiments: Supervised linear probing on frozen ResNet-18 encoder across labeled data budgets: 10%, 20%, 40%, 60%, 80%, 100%. Cross-entropy loss, SGD optimizer (lr=0.01,	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022

		momentum=0.9). 5-fold stratified cross-validation per budget.		
15 Feb 2026	17 Feb 2026	Full supervised baseline (100% labels): Training ResNet-18 from scratch (no SSL pretraining) on 100% labeled data for controlled comparison. Analysis of supervised overfitting phenomenon - validation accuracy degradation from 82.39% (40% SSL) to 76.52% (100% supervised).	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009
18 Feb 2026	20 Feb 2026	Grad-CAM XAI integration: Implementation of Gradient-weighted Class Activation Mapping on the final convolutional block (layer4) of ResNet-18. Generation of saliency heatmaps overlaid on P-QRS-T waveform regions for all 5 arrhythmia classes. Qualitative validation with clinical literature.	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022
21 Feb 2026	22 Feb 2026	Results compilation and statistical analysis: Computation of per-class Precision, Recall, F1-Score, and AUC-ROC. Paired t-test significance testing between 40% SSL model and 100% supervised baseline. Tabulation of label efficiency curve (accuracy vs. labeled data fraction).	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009
23 Feb 2026	25 Feb 2026	Project Report Writing - Chapter Drafting: Chapters 1 (Introduction, Motivation, SDG-3 Alignment), 2 (Literature Review: SimCLR, MoCo, BYOL, ECG Digitization Methods), and 3 (System Design: Architecture Diagram, Preprocessing Pipeline Flowchart, NT-Xent Formulation).	A Lalith Rahul RA2211003011009	M V Prahlad Karthik RA2211003011022
26 Feb 2026	27 Feb 2026	Project Report Writing - Results & Discussion: Chapter 4 (Implementation: Dataset Statistics, Hyperparameter Table, Training Curves), Chapter 5 (Results: Accuracy Table, Confusion Matrices,	M V Prahlad Karthik RA2211003011022	A Lalith Rahul RA2211003011009

		Grad-CAM Heatmap Figures, Label Efficiency Analysis - 40% vs 100% Finding).		
28 Feb 2026	01 Mar 2026	Project Report Writing - Conclusion & Appendix: Chapter 6 (Conclusion, Limitations, Future Work: deployment on Android TFLite, federated learning extension), References (IEEE/ACM format), Appendix (Code Snippets, Model Architecture Diagram, Sample XAI Outputs).	A Lalith Rahul RA2211003011009 M V Prahlad Karthik RA2211003011022	Both
02 Mar 2026	03 Mar 2026	Report review, formatting, and plagiarism check. Internal viva preparation: preparation of 15-slide presentation deck (problem statement, methodology, results, demo), dry run presentation to supervisor Dr. P. Akilandeswari for feedback and corrections.	A Lalith Rahul RA2211003011009 M V Prahlad Karthik RA2211003011022	Both

Summary record of major progress meetings with supervisors (Sprint 1)

Summary record of major progress meetings with supervisors			**Working title of dissertation/research project:**	
Meeting date & supervisors present	**Progress since last meeting**	**Agreed programme of work and target dates**	**Other issues, e.g. facilities, supervision, training needs, etc.**	**Date of next meeting**
22 Jan 2026 Dr. P. Akilandeswari	Dataset acquisition complete. CLAHE preprocessing module implemented and tested on sample ECG images. GitHub	Complete adaptive thresholding for grid-line removal. Begin ResNet-18 backbone setup. Target: preprocessing	Discussed illumination variability in field-captured images. Agreed to test CLAHE clip limits 1.5, 2.0, 3.0	04 Feb 2026

	repository initialized with CI pipeline.	pipeline fully operational by 28 Jan 2026.	and select by SSIM metric.	
06 Feb 2026 Dr. P. Akilandeswari	NT-Xent loss implementation validated. ResNet-18 projection head integrated. Augmentation pipeline producing correct positive/negative pairs confirmed via visual inspection.	Begin pretraining run. Monitor NT-Xent loss convergence and t-SNE embedding quality. Target: stable pretraining loss by 09 Feb 2026.	Supervisor advised to log GPU memory usage during training and document batch size vs. memory trade-offs for report. Recommended SimCLR paper Appendix reference.	11 Feb 2026
11 Feb 2026 Dr. P. Akilandeswari	Sprint 1 complete. Pretraining converged at epoch 187. t-SNE shows clear inter-class cluster separation in 128-dim embedding space. Best checkpoint saved.	Proceed to Sprint 2: label budget fine-tuning experiments. Begin with 10%, 20%, 40% labeled subsets. Set up evaluation harness with 5-fold CV.	Sprint 1 review: all preprocessing and pretraining milestones met on schedule. No infrastructure issues. Colab Pro+ GPU quota sufficient.	18 Feb 2026

Summary record of major progress meetings with supervisors (SPRINT 2)

Summary record of major progress meetings with supervisors			**Working title of dissertation/research project:**	
Meeting date & supervisors present	**Progress since last meeting**	**Agreed programme of work and target dates**	**Other issues, e.g. facilities, supervision, training needs, etc.**	**Date of next meeting**
18 Feb 2026 Dr. P. Akilandeswari	Fine-tuning experiments across 10%-100% label budgets complete. Key finding: 40% labeled data achieves peak accuracy of 82.39%. 100% labeled data degrades to 76.52%	Implement Grad-CAM XAI saliency heatmaps on final convolutional layer. Validate heatmap alignment with P-QRS-T clinical landmarks. Document label efficiency finding as primary contribution.	Supervisor confirmed label efficiency finding is a significant and publishable result. Advised to include AUC-ROC curves and per-class F1 scores in results	25 Feb 2026

	due to supervised overfitting on limited fine-tuning samples.		chapter. Suggested IEEE JBHI as target venue.	
25 Feb 2026 Dr. P. Akilandeswari	Grad-CAM heatmaps generated for all 5 arrhythmia classes. Qualitative review confirms activation regions align with QRS morphology and ST-segment regions for respective pathologies. Statistical analysis (paired t-test) confirms significance of SSL vs. supervised baseline difference ($p < 0.01$).	Begin full report writing. Chapters 1-3 (Introduction, Literature Review, System Design) to be completed by 01 Mar 2026. Chapter 4-6 by 03 Mar 2026.	Report format guidelines reviewed against SRM Institute B.Tech project report template. IEEE citation style confirmed. Plagiarism check via Turnitin required before submission.	03 Mar 2026
03 Mar 2026 Dr. P. Akilandeswari	Draft report (Chapters 1-6) submitted for supervisor review. Presentation deck (15 slides) prepared. Plagiarism score: 7% (within permissible limit). All figures, tables, and references formatted and verified.	Incorporate supervisor feedback on Chapters 3 and 5. Final report submission by 05 Mar 2026. Prepare for internal viva on 05 Mar 2026.	Supervisor provided minor revision comments on Discussion section. Commended clarity of NT-Xent loss derivation and label efficiency analysis. No major structural changes required.	05 Mar 2026

Summary record of major progress meetings with supervisors (SPRINT 3)

Summary record of major progress meetings with supervisors			**Working title of dissertation/research project:**	
Meeting date & supervisors present	**Progress since last meeting**	**Agreed programme of work and target dates**	**Other issues, e.g. facilities, supervision, training needs, etc.**	**Date of next meeting**

05 Mar 2026 Dr. P. Akilandeswari	Final report submitted to department after incorporating all supervisor revisions. Internal viva conducted. System demo of live ECG photograph inference and Grad-CAM heatmap visualization presented to evaluation panel.	No further development tasks. Post-submission: prepare IEEE paper draft based on label efficiency finding for potential journal submission.	Internal viva evaluation panel appreciated the clinical relevance and SDG-3 alignment. Panel recommended exploring federated learning extension as future work. Grade pending official declaration.	NA - Project Complete

Worksheet / Data collection / Observation etc

PROJECT OUTCOMES AND RESULTS SUMMARY

1. PRIMARY PERFORMANCE METRIC - LABEL EFFICIENCY FINDING (KEY CONTRIBUTION):

The most significant and counter-intuitive finding of this project is the demonstration of label efficiency superiority in the self-supervised contrastive learning paradigm. The system achieved a peak diagnostic accuracy of 82.39% when fine-tuned on only 40% of the available labeled training data. When the labeled data budget was increased to 100% (i.e., all available labeled samples used for supervised fine-tuning), the classification accuracy declined to 76.52%.

This 5.87 percentage point accuracy degradation under full label supervision is attributed to the phenomenon of supervised overfitting: with a small fine-tuning dataset size, the linear classifier appended to the frozen SSL encoder memorizes training-set noise rather than generalizing the learned contrastive representation. The SSL pretraining phase has already encoded the discriminative ECG morphology features in the encoder weights; additional labeled data without regularization causes the classification head to overfit rather than improve.

This finding has direct clinical and deployment implications: it establishes that high-quality arrhythmia detection is achievable with minimal clinical annotation effort (40% of dataset), dramatically reducing the cost barrier for deploying AI-assisted cardiac screening in environments where expert ECG annotation is scarce.

2. QUANTITATIVE RESULTS SUMMARY:

- Peak Diagnostic Accuracy (40% labeled budget, SSL pretrained): 82.39%

- Baseline Supervised Accuracy (100% labeled, no SSL): 76.52%

- Label Efficiency Gain: +5.87% absolute accuracy with 60% less annotation overhead

- Arrhythmia Classes: Normal Sinus Rhythm, Atrial Fibrillation, Left Bundle Branch Block, Right Bundle Branch Block, ST-Elevation (5-class classification)

- Encoder Architecture: ResNet-18 (pretrained backbone, ImageNet weights, modified projection head: 512 -> 256 -> 128 dimensions)

- Training Protocol: NT-Xent loss, tau=0.07, batch size 256, 200 epochs pretraining, Adam optimizer (lr=3e-4)

- Explainability: Grad-CAM saliency heatmaps on Layer4 of ResNet-18, validated against clinical P-QRS-T landmark literature

3. EXPLAINABILITY AND CLINICAL TRUST:

Gradient-based saliency heatmaps generated via Grad-CAM confirm that the model attends to clinically relevant regions: QRS complex width for Bundle Branch Block classes, P-wave morphology for Atrial Fibrillation, and ST-segment elevation patterns for the ST-Elevation class. This provides physician-interpretable evidence for each diagnostic output, a critical requirement for clinical deployment and regulatory consideration.

4. SOCIETAL IMPACT AND SDG-3 ALIGNMENT:

The system requires no proprietary ECG digitization hardware, operates on commodity Android smartphones with 8MP cameras, and performs inference via a quantized ResNet-18 model (~12 MB INT8). This architecture directly enables deployment by ASHA workers and primary care physicians in Tier-2/Tier-3 districts, democratizing cardiac screening in alignment with UN Sustainable Development Goal 3 (Good Health and Well-Being) and the Ayushman Bharat Digital Mission framework.

Journal Publication Details

Status: Manuscript in preparation. Target venue: IEEE Society on Social Implications of Technology (SSIT) Conference on Sustainable AI for Social Impact and Global Development (IEEE SASIGD) .
Planned submission:May 2026.
Working Title: "Self-Supervised Contrastive Learning for Label-Efficient Arrhythmia Detection from Smartphone-Captured ECG Images with Explainable AI." Authors: A Lalith Rahul, M V Prahlad Karthik, Dr. P. Akilandeswari.